Original Research

High Rates of Return to American Football After Primary Hip Arthroscopy for Femoroacetabular Impingement

A 2-Year Minimum Follow-up

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Background: Femoroacetabular impingement (FAI) is a common cause of hip pain in young, active athletes. American football players place high demands on the hip due to the sport's requirements for pivoting, cutting, and loading in deep flexion. Although arthroscopic treatment of FAI in professional National Football League (NFL) players has shown favorable outcomes, football-specific outcomes in non-NFL athletes remain limited.

Purpose: To report the return-to-sport (RTS) rates and functional outcomes at a minimum 2-year follow-up for active American football players competing at multiple levels after primary hip arthroscopy for FAI.

Study Design: Case series; Level of evidence, 4.

Methods: An institutional database was queried to identify all active American football players who underwent primary hip arthroscopy with labral repair between 2010 and 2023. Demographic characteristics, radiographic parameters, procedure details, complications, and reoperations were collected. Follow-up was completed at a minimum of 2 years to obtain updated patient-reported outcome measures (PROMs) as well as football-specific data, including current participation and RTS status. PROMs were reported as mean ± standard deviation, and categorical variables, including RTS, were reported as frequencies. Pre- versus postoperative PROMs were compared using *t* tests or Mann-Whitney *U* tests to assess outcomes.

Results: In total, 49 hips in 38 male football players (age, 18.1 years; range, 14-26 years) were included. Most athletes played at the high school (74%) or college level (21%). A total of 51% played as linemen. Labral repair was performed in all cases. At a mean 7.6 years, mean modified Harris Hip Score was 90 ± 13, Hip Outcome Score–Activities of Daily Living was 95 ± 7, and Hip Outcome Score–Sport was 89 ± 17. Overall mean surgery satisfaction was 8.7 ± 1.7 (out of 10). Significant postoperative improvements were observed in all PROMs ($P \leq .002$). Twelve players (32%) did not attempt RTS due to other nonmedical factors and were excluded from return to sport analysis. A total of 23 (88%) players achieved RTS, and 3 players (12%) did not. Two players (8%) cited other injuries, and 1 player (4%) continued to have hip limitations. Four hips (8%) underwent reoperation at a mean 4.9 years postoperative.

Conclusion: Amateur American football players undergoing primary hip arthroscopy for FAI demonstrated excellent patient-reported outcomes and a high RTS rate of 88%, when excluding those who did not return for nonmedical reasons, at a mean 7.6-year follow-up. Notable positional differences were observed across multiple levels of play. Although outcomes were generally favorable, both medical and nonmedical factors could limit return to sport, particularly among high school and college athletes.

Keywords: football; hip; femoroacetabular impingement; arthroscopy; medical aspects of sports

Femoroacetabular impingement (FAI) is a syndrome of abnormal contact between deformities of the femoral head-neck junction (cam-type), acetabulum (pincer-type),

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or both that can cause severe hip pain and functional limitations.[27] Mechanical conflict between the femoral-sided lesion and the acetabular rim can cause labral tears.[26] The cause of FAI is linked to skeletal development, with radiographic signs seen as early as ages 10 to 12.[1,26] Biomechanical studies show increased loading in typical cam lesion areas during movements like internal rotation and flexion, suggesting that early physical activity may increase FAI risk.[6,23,26] Frank et al[8] reported a 55% prevalence of cam deformities in athletes versus 23% in the general population. Overall, FAI prevalence in athletes ranges from 64% to 89%, varying by sport and competition level.[10,22,24]

American football players place significant repetitive stress on the hip, with frequent cutting, twisting, deep hip flexion, and contact trauma. Kapron et al[9] found that 72% of 67 asymptomatic college football players had an alpha angle >50°, with 48% showing bilateral abnormalities. Sochacki et al[25] reported an 84% return-to-sport (RTS) rate among 55 NFL players after arthroscopic hip surgery for FAI, with a 1-year continued play rate of 79%. Similarly, Menge et al[14] observed an 87% RTS rate in 51 professional National Football League (NFL) players. Additionally, several studies have shown significantly higher RTS rates after hip arthroscopy in elite athletes compared with lower level athletes in the same sport.[3,19]

Few studies exist evaluating outcomes of football players after arthroscopic hip management of FAI, especially at the high school or college level. With the National Collegiate Athletic Association (NCAA) rule changes in 2021 regarding "name, image, and likeness," college football players now face greater financial incentives to protect their earning potential, which may significantly influence injury management and RTS decisions.[12] This further highlights the importance of understanding RTS outcomes after hip labral injuries in these athletes to support patient-centered decision making.

The purpose of our study was to report the RTS rates and functional outcomes of American football players across multiple levels at a minimum 2-year follow-up after primary hip arthroscopy for FAI. We hypothesized that although RTS rates would remain favorable, the overall rate would be decreased compared with previous studies reporting on professional American football players.

METHODS

Participant Characteristics

After approval from the institutional review board was granted (IRB No. 08-002259), a retrospective analysis was performed on a prospectively maintained database of patients who underwent primary hip arthroscopy between 2010 and 2023 at a high-volume academic center. Eligibility criteria included active participation in football at the time of surgery, a primary hip arthroscopy for FAI, and an intention to RTS postoperatively. Patients were excluded if they had follow-up of <2 years, underwent concomitant periacetabular osteotomy (PAO), lacked sufficient football-related documentation in their electronic medical records, or opted not to participate in the study (Figure 1).

Data Collection

A retrospective review of electronic medical records was conducted to gather preoperative patient data, including demographic variables (age, sex, body mass index [BMI], and football participation), radiographic findings (Tönnis grade, lateral center-edge angle, and alpha angle), and patient-reported outcome measures (PROMs). Postoperative variables collected included surgical details (labral repair, capsulotomy, capsular repair, cam resection), RTS, PROMs, and the necessity for revision surgery. Follow-up assessments were completed at a minimum of 2 years postoperatively using Research Electronic Data Capture (REDCap; Vanderbilt University) and telephone interviews. These follow-ups aimed to obtain updated PROMs as well as football-specific data, including player position, current level of participation, and RTS status. RTS was defined as participation in ≥1 competitive American football game—excluding practices or scrimmages—at the same or a lower level than before surgery. Outcome measures included the modified Harris Hip Score (mHHS), Hip Outcome Score (HOS) for both Activities of Daily Living (HOS-ADL) and Sport (HOS-Sport) subscales, Tegner Activity Scale, visual analog scale (VAS) for the presence of pain at rest and during activity, a 5-point VAS pain scale, and overall patient satisfaction score on a 10-point scale.

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Ethical approval for this study was obtained from Mayo Clinic, Rochester, Minnesota, USA.

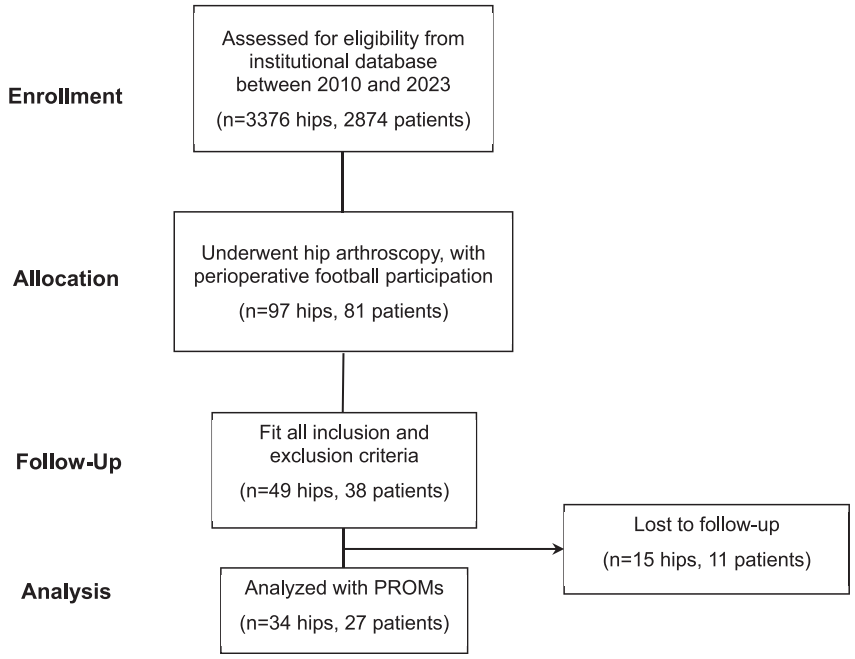

Figure 1. CONSORT (Consolidated Standards of Reporting Trials) flow diagram. PROMs, patient-reported outcome measures.

Surgical Technique

After the patient was positioned supine on a modified traction table, a padded perineal post was used to secure the patient, and distraction of the operative hip was performed. The surgical site was then prepared and draped using standard sterile technique. Standard anterolateral and midanterior portals were created. Interportal or T-capsulotomy was performed depending on surgeon preference for access. After diagnostic arthroscopy, labral repair was performed with suture anchors, alternating viewing and working portals as needed. After completion of intra-articular debridement, labral repair, and/or pincer decompression, traction was released. A distal anterolateral accessory portal was established, through which an arthroscopic shaver was introduced to debride the pericapsular fat and the ascending rectus tendon. Cam resection was performed with use of an arthroscopic bur until the cam deformity was sufficiently addressed. Capsular plication/repair was performed in all except 1 patient, who had significant preoperative capsular tightness. Additional data on operative procedures performed are provided in Table 1. All patients followed a standardized hip arthroscopy rehabilitation protocol, beginning with partial weightbearing and passive hip range of motion for the first 2 weeks, followed by progressive weightbearing and functional exercises. By 12 weeks, patients could begin jogging and plyometrics, with limb symmetry index goals of 75% at 12 weeks and 90% at 16 weeks before full RTS.

Statistical Analysis

Statistical analyses were conducted to summarize and compare study variables. Continuous data were presented as mean ± standard deviation with 95% CI, whereas categorical variables were reported as frequencies. A chi-square goodness-of-fit test assessed distributions of categorical variables. For continuous variables, parametric data were compared using independent t tests, whereas nonparametric data were analyzed with the Mann-Whitney U test. To quantify improvements in PROMs, we used thresholds for minimal clinically important difference (MCID) previously published in hip arthroscopy patients at minimum 2-year follow-up. These included 9.2 for mHHS, 9.7 for HOS-ADL, and 14.3 for HOS-Sport.[17] The RTS rate was calculated excluding athletes who did not return due to nonmedical reasons. High school and college athletes who did not return for nonmedical factors (eg, personal choice, graduation) were excluded from the RTS rate but included in the overall results to report their effect on return to football in this population. To adhere to an intention-to-treat approach, all patients were included in the comparative analysis of PROMs. Logistic regression was performed to identify factors influencing RTS. Odds ratios and corresponding 95% CIs were calculated to evaluate associations between predictors and RTS outcomes. All statistical computations were performed using R (R Foundation for Statistical Computing), with statistical significance established at $P < .05$.

RESULTS

Patient Demographic Characteristics

A total of 49 hips in 38 male football players were included in the study, with a mean age of 18.1 ± 3.0 years and a mean BMI of 28.5 ± 6.2 kg/m^2 (Table 1). The right hip was affected in 47% of cases, the left hip in 24%, and both hips in 29%. Most athletes played at the high school level (74%), followed by college athletes (21%), 1 semiprofessional (3%), and 1 professional (3%). Notably, 51% of the total cohort played as linemen, with an even higher proportion among high school players (57% linemen). Radiographic evaluation showed a mean lateral center-edge angle of 31.9° ± 7.1°, a Tönnis angle of 4.3° ± 4.2°, and an alpha angle of 59.8° ± 12.1°. Coxa profunda was observed in 2 cases, whereas no instances of protrusio acetabuli were identified. Labral repair was performed in all 49 cases (100%). Interportal capsulotomy (59%) was more common than T-capsulotomy (41%). Femoroplasty was performed in 49 hips (100%), and acetabuloplasty was performed in 42 hips (86%).

Return to Football

A total of 23 (88%) football players successfully returned to football—excluding those who did not return for nonmedical reasons (Table 2). Among them, all resumed play at the same or a higher level compared with their preoperative participation. By the final follow-up, only 1 individual (4%) remained actively competing. In total, 15 athletes did not return; however, 12 (32%) of these cited nonmedical reasons for not returning (eg, personal choice, graduation) and were excluded from the RTS calculation. They are, however, included in Figure 2 to illustrate their effect on overall return rates in this population, with reasons such as time constraints, changing priorities, graduation, shifting focus to another sport, and financial limitations. Of the 3 athletes who tried to return but could not, 2 athletes (8%) did not return due to other injuries and 1 athlete (4%) continued to have hip limitations. Considering those who cited nonmedical reasons, the overall RTS rate was 61%.

Factors Influencing Return

To identify predictors of return to football, a logistic regression model was applied (Table 3). The variables assessed included BMI (<25 vs ≥25 kg/m^2), age (<20 vs ≥20 years), level of play (high school vs college), alpha angle (>60° vs ≤60°), and player position (nonlineman vs lineman). None of these factors demonstrated a statistically significant association with RTS.

Patient-Reported Outcomes

The mean follow-up period for this cohort was 7.6 ± 3.9 years. The mean mHHS at follow-up was 90.2 ± 12.6 (95% CI, 85.2-95.1), whereas the mean HOS-ADL and HOS-Sport were 94.5 ± 6.8 (95% CI, 91.8-97.2) and 88.7 ± 17.0 (95% CI, 85.0-95.4), respectively (Table 4). The mean postoperative Tegner activity level was 6.0 ± 2.3 (95% CI, 5.9-7.8). Among respondents on the VAS pain 5-point scale, 16 athletes (64%) reported their hips were much better than before surgery, and 8 athletes (32%) noted slight improvement. Surgery satisfaction was

TABLE 1
Patient Demographic Characteristics and Surgical Characteristics of American Football Players[a]

Characteristic	Value	P
Age, y	18.1 ± 3.0	
Patients, n (all male)	38	
Body mass index, kg/m^2	28.5 ± 6.2	
Laterality		.131
Left	9 (24)	
Right	18 (47)	
Bilateral	11 (29)	
Level of play		<.001
High school	28 (74)	
College	8 (21)	
Semiprofessional	1 (3)	
Professional	1 (3)	
Position		
Lineman	19 (51)	
Nonlineman	18 (49)	
Radiographic measures		
Lateral center edge angle, deg	31.9 ± 7.1	
Tönnis angle, deg	4.3 ± 4.2	
Alpha angle, deg	59.8 ± 12.1	
Coxa profunda	2 (5)	
Operative procedures		
Labral repair	49 (100)	
Capsulotomy		
Interportal	29 (59)	
T-capsulotomy	20 (41)	
Femoroplasty	49 (100)	
Acetabuloplasty	42 (86)	
Capsular repair	48 (98)	

[a]Values are displayed as mean ± SD or n (%).

TABLE 2
Return-to-Sport Outcomes in American Football Players After Hip Arthroscopy (n = 26)[a]

Variable	Result
Returned to football	
Yes	23 (88)
No	3 (12)
Returned at same or higher level	23 (100)
Currently active	1 (4)
Reasons for not returning	
Hip limitations	1 (4)
Other injuries	2 (8)

[a]Values are displayed as n (%). After exclusion of those who did not return due to graduation, changing priorities, or other nonmedical reasons, n = 26 patients.

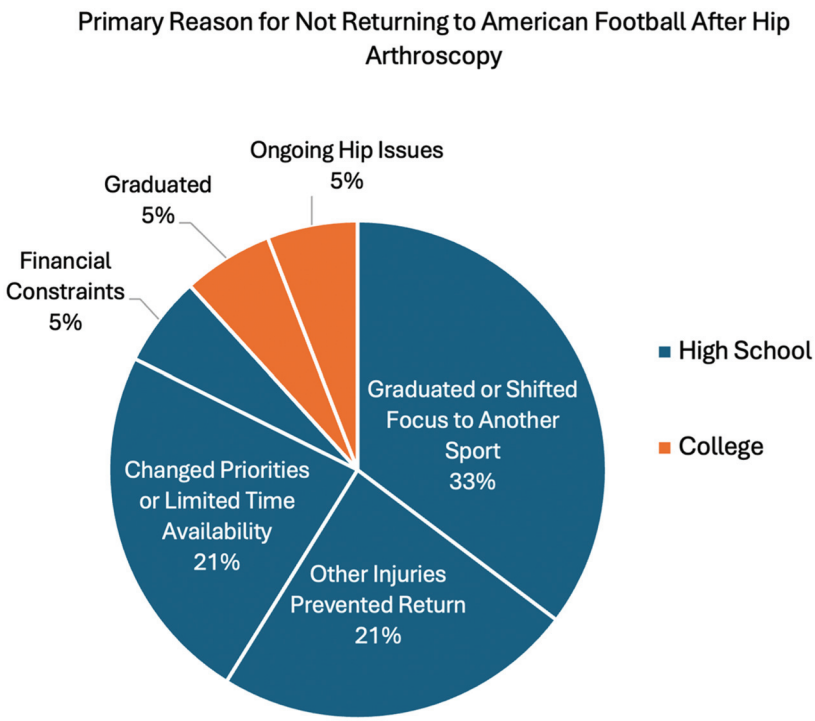

Figure 2. All reported reasons for not returning to American football after hip arthroscopy.

TABLE 3
Logistic Regression Results for Predictors of Return to American Football After Hip Arthroscopy

Variable	Odds Ratio	95% CI	*P*
Body mass index (<25 vs ≥25 kg/m^2)	1.42	0.33-5.95	.63
Age (<20 vs ≥20 y)	0.65	0.12-2.99	.59
Level of play (high school vs college)	0.29	0.04-1.43	.16
Alpha angle (>60° vs ≤60°)	1.86	0.51-7.07	.35
Football position (nonlineman vs lineman)	1.95	0.53-7.63	.32

TABLE 4
Postoperative Patient-Reported Outcome Measures in All American Football Players at Final Follow-up[a]

Variable	Value	95% CI
Mean follow-up, y	7.6 ± 3.9	
Modified Harris Hip Score	90.2 ± 12.6	85.2-95.1
Hip Outcome Score–Activities of Daily Living	94.5 ± 6.8	91.8-97.2
Hip Outcome Score–Sport	88.7 ± 17.0	82.0-95.4
Tegner activity level before surgery	8.4 ± 1.6	7.8-9.1
Tegner activity level after surgery	6.0 ± 2.3	5.9-7.8
VAS pain at rest	1.2 ± 1.7	0.5-1.8
VAS pain with use	2.2 ± 2.7	1.1-3.2
VAS pain 5-point scale		
Much worse than before surgery	0	
Slightly worse than before surgery	0	
No change	1	
Slightly better than before surgery	8	
Much better than before surgery	16	
Surgery satisfaction	8.7 ± 1.7	8.0-9.4

[a]Values are displayed as mean ± SD or n. VAS, visual analog scale.

a mean of 8.7 ± 1.7 (95% CI, 8.0-9.4). Comparing preoperative and postoperative scores, we observed significant improvements in mHHS (68.5 vs 91.0), HOS-ADL (73.9 vs 94.5), and HOS-Sport (49.7 vs 88.7) ($P \leq .002$ for all) (Figure 3). Mean improvement in scores exceeded the MCID for mHHS (9.2), HOS-ADL (9.7), and HOS-Sport (14.3) when compared with preoperative values. Postoperative outcome comparisons revealed no statistically significant differences between athletes who returned to football and those who did not return (Table 5).

Complications and Reoperations

No athletes had any perioperative complications, other than the subsequent indications for reoperation. Four hips (8%) in 4 athletes underwent reoperation for any reason at a mean 4.9 ± 4.6 years (range 0.8-12.5 years) postoperative. There were 2 revision hip arthroscopies (4%). One patient underwent revision labral repair for residual cam lesion and heterotopic ossification (HO) excision. The other patient underwent revision labral debridement, pincer resection, and concomitant PAO. The remaining 2 reoperations included subsequent PAO and isolated HO excision.

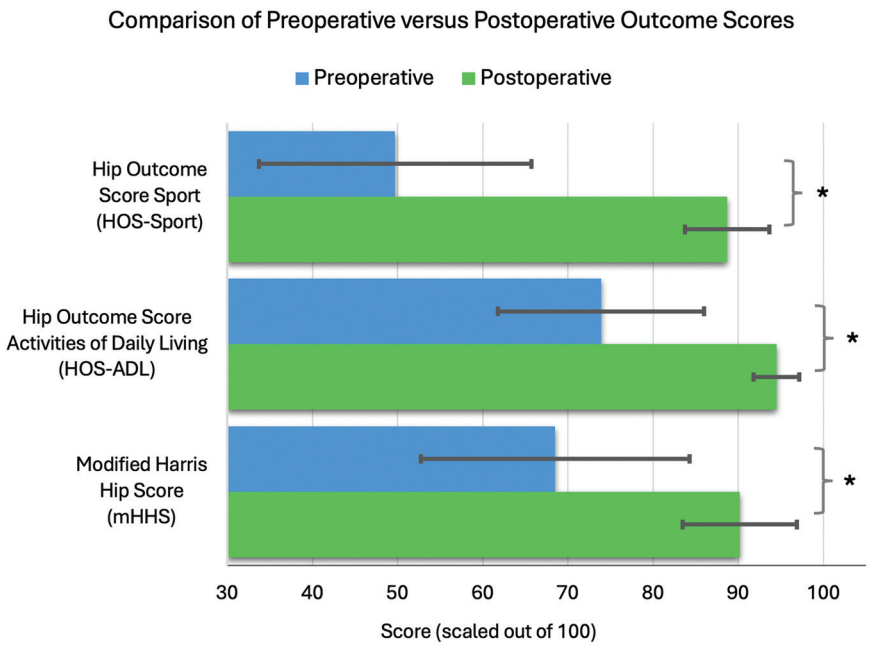

Figure 3. Preoperative versus postoperative outcome scores in American football players undergoing primary hip arthroscopy.

TABLE 5
Comparison of Postoperative Patient-Reported Outcome Measures in American Football Players Who Returned to Sport Versus Those Who Did Not[a]

Variable	Returned to Sport (n = 23)	Did Not Return (n = 15)	*P*
Modified Harris Hip Score	89.1 ± 14.8	92.3 ± 6.2	.482
Hip Outcome Score–Activities of Daily Living	95.7 ± 5.3	92.2 ± 9.0	.174
Hip Outcome Score–Sport	89.2 ± 18.8	87.8 ± 13.5	.429
Tegner activity level before surgery	8.6 ± 1.6	8.2 ± 1.6	.499
Tegner activity level after surgery	6.9 ± 2.3	6.8 ± 2.3	.916
VAS pain at rest	1.1 ± 1.4	1.3 ± 2.4	.889
VAS pain with use	2.2 ± 2.6	2.1 ± 2.9	>.999
VAS pain 5-point scale			.477
Much worse than before surgery	0	0	
Slightly worse than before surgery	0	0	
No change	1	0	
Slightly better than before surgery	3	5	
Much better than before surgery	5	11	
Surgery satisfaction	8.9 ± 0.9	8.6 ± 2.1	.649

[a]Values are displayed as mean ± SD or n. VAS, visual analog scale.

DISCUSSION

The major findings in our study demonstrated that American football players undergoing primary hip arthroscopy for treatment of FAI reported an RTS rate of 88%, excluding those who did not return for nonmedical reasons. At a mean follow-up of 7.6 ± 3.9 years, the cohort demonstrated significant sustained improvements in patient-reported outcomes. No significant differences were observed in any outcomes between football players who achieved RTS and those who did not, with comparable mHHS (89 vs 92, respectively), HOS-ADL (96 vs 92), HOS-Sport (89 vs 88), VAS pain at rest (1.1 vs 1.3), VAS pain with use (2.2 vs 2.1), and patient satisfaction (8.9 vs 8.6). The reoperation rate was 8%, and no postoperative complications were reported. We found no significant differences in RTS when comparing age, position, or high school to college.

The RTS rate in football players after arthroscopic surgery for FAI has been studied mainly in professional NFL players. Menge et al[14] found an 87% RTS rate in 51 professional players in 2017, whereas Nwachukwu et al[18]

reported a 93% RTS rate at a mean of 6 months in 40 NFL athletes 2018. In 2019, Sochacki et al[25] examined 55 NFL players (63 hips) and reported an 84% RTS rate, with 79% still playing at 1-year follow-up, and a career length (3.5 ± 2.1 years) similar to matched controls. Our RTS rate of 88% aligns with these findings, although our cohort—primarily high school and college athletes (age 18 ± 3 years)—was notably younger than the professional cohorts assessed by Menge et al (27 ± 4 years) and Sochacki et al (28 ± 3 years). In our study, the overall RTS rate of 61% including players who did not return for nonmedical reasons demonstrates the nuance and intricacy of managing symptomatic FAI in younger athletes. Many high school and college athletes cited changing priorities or graduating as reasons for not returning. Consistent with this, players who did not return for nonmedical reasons were excluded from our RTS rate, following the approach used in previous studies by Daruwalla et al[5] and Bono et al,[2] which similarly excluded collegiate athletes unable to return due to nonmedical factors. Consideration of these variables and the patient's individual goals can help providers tailor management of FAI and counsel patients appropriately to optimize outcomes.

At a mean 7.6 ± 3.9 years of follow-up, we reported mean scores for mHHS (90.2), HOS-ADL (94.5), HOS-Sport (88.7), VAS pain at rest (1.2/10), VAS pain with use (2.2/10), and surgery satisfaction (8.7/10). These findings support the efficacy and midterm durability of hip arthroscopy symptomatic FAI. Domb et al[7] reported similar minimum 2-year follow-up PROMs of 86.1 for mHHS, 86.9 for Nonarthritic Hip Score, 88.5 for HOS-ADL, 76 for HOS-Sport, 1.6 for VAS, and 8.5 for surgery satisfaction in athletes of various sports, including football, although only 61% underwent labral repair compared with 100% in our cohort. Our slightly higher PROMs may reflect our younger (mean 18 vs 31 years) and exclusively male population, as both age and sex are known outcome predictors.[7,11,13] The minimal PROM differences between those who did and did not RTS in our study highlight the role of nonmedical factors influencing return at the high school and collegiate levels.

A sport-specific consideration for football players with symptomatic FAI is position, as BMI and physical build vary significantly depending on the player's role. Consistent with findings in NFL players from Menge et al[14] (59%) and Sochacki et al[25] (32%), linemen comprised the largest subgroup in our cohort, representing 51% of players who reported position and 57% of high school athletes. Higher BMI, common among linemen, may increase hip compressive and shear forces, contributing to symptomatic FAI, as supported by epidemiological data showing that 42% of FAI patients are overweight or obese.[4] Furthermore, the crouched 3-point stance typical of linemen subjects the hip to maximal flexion and asymmetric loading, increasing stress and risk of labral tears. Although hip arthroscopy remains effective regardless of BMI, higher BMI has been linked to lower PROMs.[20,21] The logistic regression conducted in this study did not show a significant relationship between BMI (<25 vs ≥25 kg/m^2) and RTS.

Primary hip arthroscopy for FAI has demonstrated efficacy with favorable reoperation and revision rates. A meta-analysis by Minkara et al,[16] including nearly 2000 hips (29% athletes), reported a pooled reoperation rate of approximately 6%, with revision arthroscopy accounting for 1%. Another meta-analysis found no significant difference in revision rates between athletes and nonathletes (3% vs 2%).[15] Football-specific studies report revision rates of 3% to 8% in NFL players.[14,25] Our cohort showed an 8% reoperation rate and a 4% revision rate, aligning with existing data. These findings suggest that revision rates for competitive high school and collegiate football players undergoing primary hip arthroscopy for FAI are comparable with those seen in other athletes and young adults, although further research is needed.

The present study is not without limitations. PROMs were collected from 27 of the 38 athletes (71%), which may introduce response bias and allow outliers to disproportionately influence the mean scores. Additionally, the 12 athletes who did not return for nonmedical reasons could bias the results. The regression model for RTS predictors should be interpreted cautiously given the limited sample size. Furthermore, procedures were performed by multiple surgeons over 13 years (2010-2023), during which advancements in hip arthroscopy may have led to variations in FAI management and patient outcomes.

CONCLUSION

Amateur American football players undergoing primary hip arthroscopy for FAI demonstrated excellent patient-reported outcomes and a high RTS rate of 88%, when excluding those who did not return for nonmedical reasons, at a mean 7.6-year follow-up. Notable positional differences are observed across multiple levels of play. Although outcomes are generally favorable, both medical and nonmedical factors can still limit RTS, particularly among high school and college athletes.

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